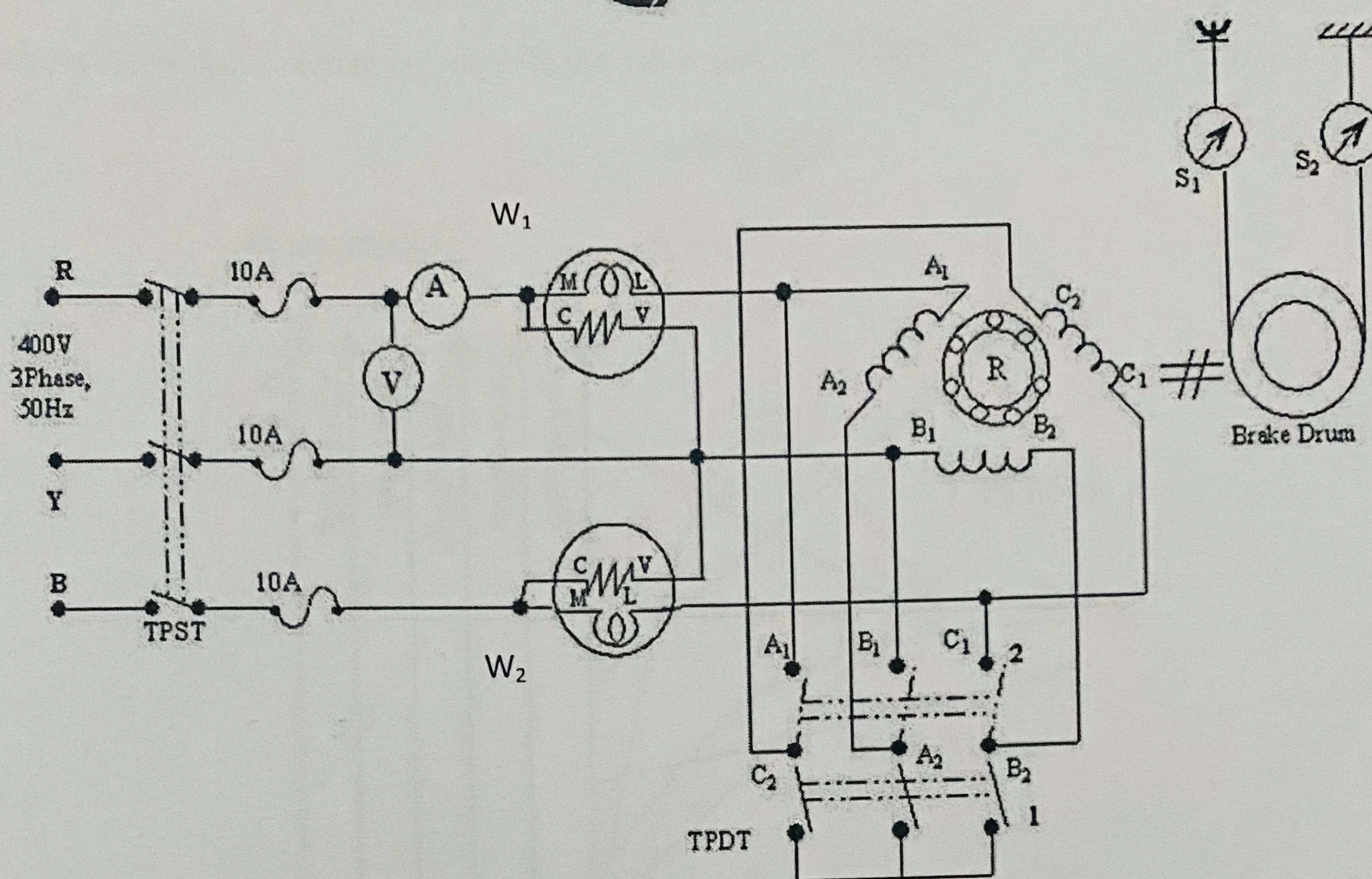




SAMPLE CALCULATION (Set No.)

Radius of brake drum $R = \dots\dots\dots m$

Synchronous speed $N_s = \dots\dots\dots rpm$

Input Power $W = W_1 + W_2 = \dots\dots\dots W$

Power factor $\cos\phi = \cos[\tan^{-1} \frac{\sqrt{3} \times (W_1 - W_2)}{(W_1 + W_2)}] = \dots\dots\dots$

Percentage slip, $s = \frac{N_s - N}{N_s} \times 100 = \dots\dots\dots \%$

Torque $T = R \times (S_1 - S_2) \times g = \dots\dots\dots N-m$

Output power $= \frac{2\pi NT}{60} = \dots\dots\dots W$

Efficiency $= \frac{\text{Output}}{\text{Input}} \times 100 = \dots\dots\dots \%$

Additional KVAR required to improve the power factor to 0.95 ($\cos\phi_2$) =

$= W(\tan\phi_1 - \tan\phi_2) \times 10^{-3} = \dots\dots\dots$